Activity 8 - Develop a Renewable-Energy Business Case

Alignment: K4 | A6 | LO3 Duration: 55 minutes Objective: Compare efficiency, electrification and renewable-energy options using emissions integrity, economics and implementation risk.

Scenario

EcoForge uses 4.2 GWh of grid electricity and natural gas process heat. Its roof can host 650 kWp solar, one oven may be electrified, chilled-water optimisation promises savings, and suppliers offer renewable-electricity contracts and certificates. Capital is constrained and the lender requires transparent emissions treatment.

Materials

- Hourly/annual energy profile
- Illustrative capex, opex and tariff assumptions
- Grid-factor and certificate integrity prompts
- Business-case workbook

Questions

- Which option reduces demand before changing supply?
- What emissions accounting and claim conditions apply?
- What are NPV, payback and cost per tCO2e?
- Which risks and sensitivities could reverse the recommendation?

Detailed procedure

1. Establish the energy baseline, load profile, emissions basis and business constraints.
2. Model efficiency and operational controls before new supply options.
3. Estimate onsite solar generation, self-consumption, export and annual emissions effect.
4. Evaluate process electrification including load, grid trajectory, capex and operational dependencies.
5. Assess renewable-electricity procurement and certificate quality, geography, vintage, exclusivity and retirement.
6. Calculate payback, NPV, marginal abatement cost and key sensitivity cases.
7. Score delivery, technology, regulatory, data and claim risks.
8. Recommend a staged portfolio with trigger points rather than one isolated project.
9. Prepare a one-page decision brief separating inventory effect, avoided emissions and claims.

Required evidence

- Renewable-energy scenario model
- Option scorecard
- Sensitivity analysis
- Executive recommendation

Acceptance check

The recommendation uses consistent assumptions, separates demand and supply levers, quantifies economics and emissions, and states certificate and claim safeguards.

Files

- Activity-08-Workbook.xlsx - editable working file
- Activity-08-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.